IET Image Processing

WILEY

The Institution of Engineering and Technology

ORIGINAL RESEARCH OPEN ACCESS

Research on Facial Expression Recognition Method Based on Improved ConvNeXt

Dan Chen[1] | Yu Cao[1] | Xu Cheng[2]

[1]School of Information Science and Control Engineering, Liaoning Petrochemical University, Fushun, China | [2]School of Economics and Management, Shenyang Agricultural University, Shenyang, China

Correspondence: Yu Cao (caoyu@lnpu.edu.cn)

Received: 20 August 2024 | **Revised:** 7 May 2025 | **Accepted:** 22 May 2025

Funding: This work was supported in part by the Applied Basic Research Program of Liaoning Province under Grant 2022JH2/101300272, in part by the National Natural Science Foundation of China under Grant 71902121 and in part by the Applied Basic Research Program of Liaoning Province (Key Technology Research on Non-standard Agricultural Product Quality Grading Based on Artificial Intelligence Technology).

ABSTRACT

Advanced facial expression recognition technology can significantly enhance human-computer interaction and improve intelligent services for humans. This paper introduces a novel facial expression recognition method utilizing an enhanced ConvNeXt network. By integrating the SENET attention mechanism into the ConvNeXt block, key feature information extraction is effectively enhanced. Additionally, the incorporation of the focal loss (FL) function optimizes the classification performance of the network model. Experimental results show that the improved ConvNeXt network achieves higher accuracy compared to other deep learning models, with accuracy rates of 83.8% and 70.4% on the RAF-DB and FER2013 datasets, respectively.

1 | Introduction

Facial expressions play a key role in communication, expressing a wide range of emotions and complexities. Advanced facial expression recognition technology is the key to enhanced human-computer interaction. The technology has been widely applied in various fields such as human-computer interaction devices, safe driving, and medical monitoring. In human-computer interaction devices, facial expressions help to recognize the user's emotional state, customize emotional feedback, and enhance the overall experience. In the context of safe driving, monitoring the driver's facial expressions can help assess their driving status and prevent accidents caused by fatigue or emotional distress. In smart healthcare, remote monitoring of patients' facial expressions enables doctors to gain real-time insights into their physical and mental health, leading to timely therapeutic interventions.

Early research in facial expression recognition focused on capturing expression features using traditional manual techniques and combining them with classification networks. Commonly used traditional feature extraction algorithms include LBP, HOG, and sparse learning, among others. Shan et al. [1] applied LBP to extract collaborative features from adjacent pixels for facial expression recognition. Déniz et al. [2] proposed an algorithm using HOG at different scales for face recognition. Commonly used classification methods like SVM, HMM, and Bayesian classifiers are then applied to classify the feature matrix and accurately determine facial expressions. However, these methods often face challenges such as low recognition accuracy and slow processing speed in complex environments.

In recent years, convolutional neural networks (CNN) have become a prominent approach in deep learning for facial expression recognition. With the continual advancements and enhancements in CNNs, various algorithms have been developed, such as AlexNet, VGG16, MobileNet, ResNet, GoogLeNet, DenseNet, SHCNN, Swim Transformer, and ConvNeXt. Atabansi et al. [3] utilized a VGG-16 network model, Li et al. [4] implemented the ResNet-50, and Lei J. Y. [5] combined multi-scale dilated convolutions and DenseNet for enhanced feature extraction. Other researchers, such as Sun et al. [6], Nan et al. [7], and Li et al. [8], explored different CNN models like GoogLeNet, MobileNet, pACNN, and gACNN for facial expression recognition. Miao et al. [9] focused on analysing the SHCNN structure to address over fitting in small datasets. Furthermore, Bie et al. [10] used the Swim Transformer network applied in facial expression recognition, which is a hierarchical transformer structure and introduces a sliding window mechanism. However, some network models usually do not consider the channel equalization problem, ignoring the variability of the contribution of different channels to the task, which affects the ability of the model to capture important features.

As deep learning techniques continue to improve, researchers improve and innovate networks based on the original model, demonstrating superior performance. Nie H. [11] demonstrated that integrating SENET into the VGGNet-16 network led to higher expression recognition rates. Zhang K. [12] proposed an efficient adaptive transfer network (EATN) for aspect-level sentiment analysis. It designed a new MLP module to further enhance the transferability characteristics of deep neural networks. Yu Y. [13] introduced a variable-degree MGDRS-based MLFS algorithm that dynamically adjusts the variable-degree parameter between the pessimistic and optimistic degrees of MDGRS to accommodate different types of multi-labelled datasets, which improves the reliability of the feature selection process. Li X. [14] proposed a fusion innovative approach combining the ConvNeXt network structure and the multi-scale dilated attention (MSDA) mechanism, which comprehensively extracts fault features.

ConvNeXt combines the efficiency of CNN and the global modelling ability of the transformer, and has strong feature extraction ability, so ConvNeXt is selected as the base network in this paper. In order to further improve the accuracy of the facial expression recognition algorithm, the original ConvNeXt network is improved in this paper. The improved ConvNeXt network adds the SENET attention mechanism on the basis of ConvNeXt-T to solve the channel equalization problem. In addition, the FL function is used to alleviate the problems of category imbalance and the learning efficiency of difficult and easy samples. The experimental results show that the improved method proposed in this paper has high reliability and effectiveness.

Our main contributions can be summarized as follows:

- We propose an improved the ConvNeXt network model, which is refined on the basis of ConvNeXt-T structure. Extensive validation on facial expression recognition tasks demonstrates its superior performance, achieving promising results.
- The SENET attention mechanism is introduced in the ConvNeXt block to reduce the influence of non-expression regions and enhance the extraction of key information.
- The FL function is used to solve the problem of an unbalanced distribution of facial expression samples, which improves the classification performance of the network.
- The proposed method is effectively validated on the RAF-DB and FER2013 facial expression datasets with accuracies of 83.8% and 70.4%, respectively, and the results outperform the original ConvNeXt model and other deep learning models.

2 | Related Work

The ConvNeXt network model is designed with a modern convolutional architecture to effectively capture facial expression features. However, the model is more inclined to focus on local facial regions and may ignore the synergistic expression features in secondary regions, resulting in limited generalization ability to some micro-expressions. In addition, the traditional cross-entropy (CE) loss function causes the model to over fit high-frequency simple samples and fails to balance the learning intensity of multi-category samples. In order to improve the above problems of the model, relevant research work is carried out on both the attention mechanism and the loss function.

2.1 | Attention Mechanism

Convolutional block attention module (CBAM) uses two-dimensional modules, channel and spatial, to optimize both the channel weights and space of the feature map. However, the attention is computed through two times, which leads to high computational complexity and a large number of parameters.

Efficient channel attention (ECA-Net) is used to capture adjacent channel correlations using adjustable-size convolutional kernels ($k = 3 \sim 7$) for dynamic local modelling. However, this attention mechanism has insufficient ability to capture global information, which affects the recognition of complex facial expressions.

Squeeze-and-excitation networks (SENet) is lightweight channel modelling that dynamically suppresses extraneous channel noise through the Squeeze-Excitation mechanism. Its lightweight design outperforms CBAM and ECA. SENet are able to effectively model the dependencies between distant channels and avoid the limitations of local interactions in ECA. The inference speed is faster than that of CBAM due to the absence of the complex computation of the spatial attention module.

2.2 | Loss Function

Class balancing loss (CB-Loss) is a reweighting loss based on the effective number of samples to alleviate the class imbalance problem. However, it relies on the effective number of samples to calculate and has a high computational overhead.

FL adopts a dynamic difficult-sample focusing mechanism, which can solve the difficult–easy sample imbalance problem. By dynamically reducing the weight of easy to categorize samples, more attention is paid to difficult to categorize samples. Its inference speed is close to the standard CE, and the training time is less than that of CE-Loss, which is a lightweight loss function.

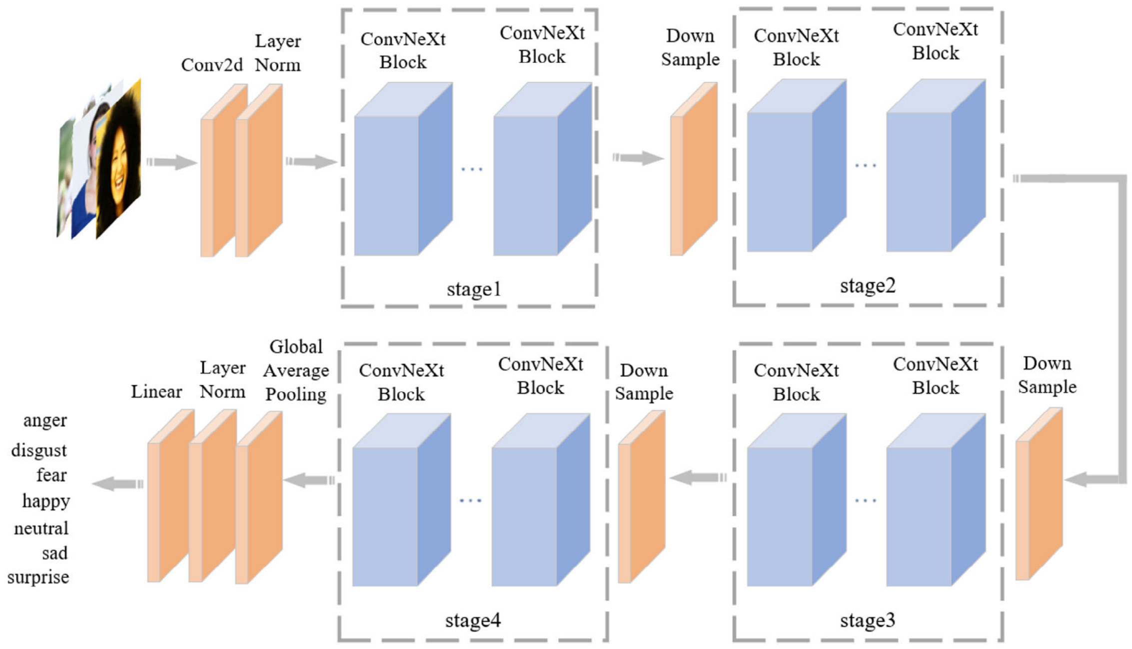

FIGURE 1 | Network structure of ConvNeXt.

3 | Methods

As a novel convolutional neural network, the ConvNeXt model has demonstrated superior recognition accuracy and computational efficiency when compared to traditional CNNs. However, there may be limitations in effectively extracting facial expression features when using ConvNeXt for facial expression recognition. To improve identification accuracy, this study proposes an enhanced ConvNeXt network model. This model integrates the SENET attention mechanism to improve feature extraction capabilities and incorporates FL to address imbalanced sample distribution in the dataset by balancing loss weights across different samples. The paper outlines the main functional blocks and provides detailed instructions for enhancing the ConvNeXt network.

3.1 | ConvNeXt Block

The ConvNeXt network model, proposed in the paper 'A ConvNet for the 2020s' published in 2022, is a pure convolutional neural network that builds upon the ResNet-50/200 model by integrating concepts and methodologies from the Swin Transformer.

Figure 1 illustrates the ConvNeXt structure. Initially, the input image undergoes down sampling through a convolutional layer with a 4×4 kernel size, reducing the image to a quarter of its original size. The downscaled image then progresses through four stages (stage1, stage2, stage3, and stage4), each containing a sequence of ConvNeXt Blocks. The down sampling in the ConvNeXt structure is accomplished by incorporating a normalized convolutional layer with a 2×2 kernel size and a stride of 2. Finally, the output is generated by employing global average pooling and fully connected layers, facilitating facial expression classification on the input image.

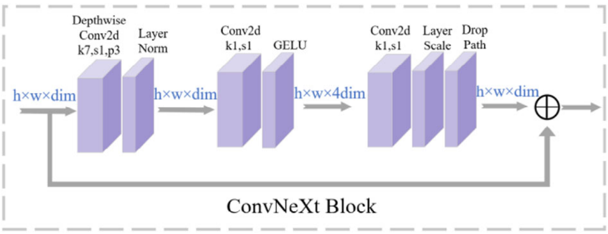

FIGURE 2 | Structure of ConvNeXt block.

The ConvNeXt block model follows a specific flow: Initially, the feature layer undergoes depth-wise separable convolution (Depwise Conv2d) using a 7×7 kernel size, one stride, and three padding. Subsequently, the feature layer, post layer normalization, attains a size of $h \times w \times$ dim. Then, a feature layer of $h \times w \times$ 4dim is generated via a convolution layer with a 1×1 kernel size, one stride, and the GELU function. This is followed by another pass through the convolutional layer, along with the layer scale and drop path. After these operations, the image is restored to $h \times w \times$ dim. Lastly, the output is combined with the input data, resulting in the final ConvNeXt block outcome. For a visual representation of the ConvNeXt block model, refer to Figure 2.

Based on varying computational complexities, ConvNeXt introduces five versions: ConvNeXt-T, ConvNeXt-S, ConvNeXt-B, ConvNeXt-L, and ConvNeXt-XL (see Table 1). Each version comprises four stages, with differences in the number of channels and ConvNeXt blocks.

The ConvNeXt network integrates features from popular backbone networks with optimized improvements. In contrast to models like VGG and inception, ConvNeXt incorporates larger kernel sizes and group convolutions. Additionally, the ConvNeXt block employs a reverse-bottleneck design concept to decrease the overall network's floating-point operations (FLOPs).

TABLE 1 | Configuration of ConvNeXt versions.

Version	Channel	Block
ConvNeXt-T	96, 192, 384, 768	3, 3, 9, 3
ConvNeXt-S	96, 192, 384, 768	3, 3, 9, 3
ConvNeXt-B	128, 256, 512, 1024	3, 3, 9, 3
ConvNeXt-L	192, 384, 768, 1536	3, 3, 9, 3
ConvNeXt-XL	256, 512, 1024, 2048	3, 3, 9, 3

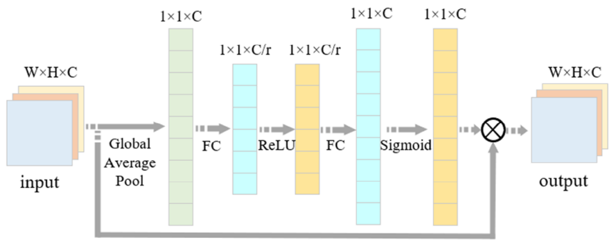

FIGURE 3 | Structure of SENET Block.

3.2 | SENET Block

SENET is a channel attention mechanism that learns the correlation between different features based on the importance of each channel and generates weight parameters for each channel. The goal is to improve the extraction of key information and reduce the impact of irrelevant features. See Figure 3.

The essence of SENET lies in its squeeze and excitation operations. The Squeeze operation utilizes a global average pooling technique to compress features across spatial dimensions. For instance, if the original feature layer U has dimensions $H \times W \times C$, post-squeeze operation, the feature layer shrinks to $1 \times 1 \times C$, expanding the perceptual field. The formula for this compression is depicted below, with Z representing the compressed feature map.

$$Z = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} U(i, j) \tag{1}$$

The excitation operation involves adding a fully connected layer after obtaining the compressed result Z from the squeezing operation to predict the importance of each channel and improve the model's generalization ability. Initially, a dimension-reduction operation is conducted using the first FC layer, with a reduction ratio of r, transforming the feature layer's dimension to $1 \times 1 \times C/r$. Subsequently, the ReLU function enhances the network's nonlinear fitting capability. Following this, an expansion operation is performed using the second FC layer, restoring the feature layer's original dimension to $1 \times 1 \times C$. Finally, the sigmoid function is applied to determine the importance S of different channels. The corresponding formula is presented below, where δ represents the ReLU function, FC denotes fully connected, and σ signifies the sigmoid function.

$$S = \sigma(\mathrm{FC}(Z, W)) = \sigma(W_2 \delta(W_1 Z)) \tag{2}$$

Various channel weights are calculated to capture the relationships between channels. These attention weights are then multiplied by the original features to amplify relevant channels and diminish irrelevant ones.

3.3 | FL Block

FL, introduced by Lin et al. in 2017, aims to address the issue of imbalanced positive and negative sample distributions. It is an optimization method that builds upon the CE loss function, allowing for flexible balancing of loss weights across different samples. While the CE loss function measures the proximity between predicted and expected probabilities, it does not account for the imbalance between easy and difficult samples. FL adjusts for this imbalance by incorporating a modulating factor γ. This factor is used to increase the weight of difficult samples and decrease the weight of easy samples, ensuring that the model does not become biased towards easier samples. The formula for FL involves adjusting the weights based on the modulating factor γ, with higher values leading to greater attenuation.

The value of γ determines the degree of attenuation, with higher values leading to higher attenuation. The formula is as follows:

$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \log_e(p_t) \tag{3}$$

Additional parameters α_t are introduced in FL to balance the weights of positive and negative samples. In binary classification, when the proportion of simple samples is large, the training process tends to prioritize these easy samples, potentially overlooking the more challenging ones and resulting in increased loss. The formula for FL is shown below:

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log_e(p_t) \tag{4}$$

For multiclass scenarios, the formula is shown below, where α_c denotes the weight of the C class sample, and p_c denotes the probability value of the C class sample obtained from the softmax output.

$$\mathrm{FL}_{soft\max} = -\alpha_c (1 - p_c)^{\gamma} \log_e(p_c) \tag{5}$$

3.4 | Improved ConvNeXt Network

In this study, the improved ConvNeXt network, referred to as ConvNeXt-T, is built upon the original ConvNeXt model. The performance of the base model is improved by incorporating the SENET attention mechanism, which assigns varying weights to each channel feature to prioritize important information.

The network architecture comprises four stages, with each stage containing 3, 3, 9, and 3 ConvNeXt blocks, respectively. The input feature layer of each stage has channel sizes of 96, 192, 384, and 768. The process initiates with the input image being pre-processed to a size of 224 × 224 × 3. After a convolutional operation with a 4 × 4 kernel size and stride of 4, followed by normalization, the image is resized to 56 × 56 × 96. Subsequently, the image goes through stage 1, which consists of 3 ConvNeXt blocks, resulting in an output image size of 56 × 56 × 96. A

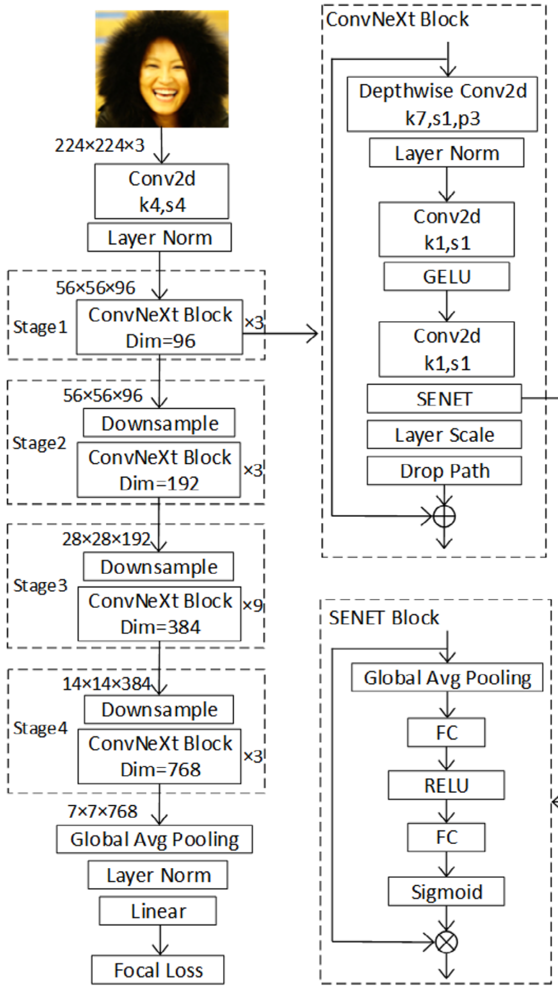

FIGURE 4 | Structure of improved ConvNeXt network.

down-sampling operation is then applied, followed by stage two with 3 ConvNeXt blocks, resulting in an output image size of 28 × 28 × 192. Another down sampling operation is carried out, followed by stage three with 9 ConvNeXt Blocks, leading to an output image size of 14 × 14 × 384. The process continues with a downsampling operation and stage 4, which includes 3 ConvNeXt blocks, resulting in an output image size of 7 × 7 × 768. Finally, the output is obtained after global average pooling, normalization, and linear operation in the fully connected layer.

The enhanced approach proposed in this study involves integrating SENET after the third convolution layer within the ConvNeXt block module across four stages. This modification aims to focus the network on crucial areas, suppress irrelevant features, and enhance the extraction of key features. Additionally, the loss function employs FL during model training to encourage the network to produce more balanced attention weights and improve generalization capabilities. The refined ConvNeXt network structure is illustrated in Figure 4.

4 | Experimental Results and Analysis

The improved ConvNeXt network underwent thorough validation on the RAF-DB and Fer2013 facial expression datasets.

FIGURE 5 | RAF-DB images.

The experimental environment, parameter configuration, and datasets are described first. Furthermore, the experimental results are meticulously analysed through two perspectives: comparison with other advanced models and ablation experiments.

4.1 | Preparation for Experiment

Experimental environment: The hardware setup for the experiment included a 13th Gen Intel(R) Core(TM) i9-13900HX 2.20 GHz processor, 16 GB RAM, Windows 11 operating system, and an NVIDIA GeForce RTX 4060 laptop GPU. The software environment utilized PyTorch 1.11.0 framework and Python 3.9.

Parameter configuration: The experiment set the initial learning rate at 0.0005, batch size at 8, and number of iterations at 50.

Data preprocessing: Due to the imbalance of data distribution in the original dataset, training data enhancement is required before training the expression recognition network. Random rotation, random erasure, random cropping, and mirroring are used for pre-processing.

4.2 | Facial Expression Datasets

4.2.1 | RAF-DB Facial Expression Dataset

RAF-DB is a database containing around 30,000 annotated facial images capturing human expressions in real-world situations. Some of these images, are shown in Figure 5. Unlike controlled laboratory datasets, RAF-DB offers a level of authenticity and diversity. This study focused on seven fundamental facial expressions, with a total of 15,339 samples analysed. Table 2 shows the distribution of different facial expression samples in RAF-DB.

4.2.2 | FER2013 Facial Expression Dataset

The Fer2013 dataset was utilized in a facial recognition challenge hosted on Kaggle in 2013. This dataset, sourced from the internet, offers a more realistic representation of real-life scenarios. The dataset includes seven expressions, and some example images are shown in Figure 6. In this study, we combined the original training set, public test set, and private test set from the FER2013 dataset, resulting in a total of 35,887 images. Table 3 lists the

TABLE 2 | Distribution of RAF-DB sample.

Class	Total
Anger	867
Disgust	877
Fear	355
Happy	5957
Neutral	3204
Sad	2460
Surprise	1619

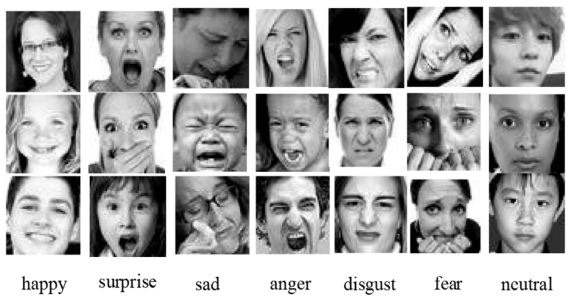

FIGURE 6 | FER2013 images.

TABLE 3 | Distribution of FER2013 sample.

Class	Total
Anger	4953
Disgust	547
Fear	5121
Happy	8989
Neutral	6198
Sad	6077
Surprise	4002

distribution of different facial expression samples in the FER2013 dataset.

4.3 | Analysis of Experimental Results

4.3.1 | Comparison With Other Advanced Models

In order to demonstrate the superiority of the proposed improved ConvNeXt network model, we conducted a comparative analysis with other state-of-the-art models. The experimental results show that the improved ConvNeXt network model achieves an accuracy of 83.8% and 70.4% on the RAF-DB, and FER2013 datasets, respectively, surpassing the performance of other existing models.

First, Table 4 gives a comparison of the accuracy of the improved ConvNeXt network proposed in this paper with other models. Obviously, the enhanced ConvNeXt network model exhibits improved classification accuracy. In addition, the method

TABLE 4 | Comparison of the accuracy of different models on the RAF-DB dataset.

Model	Accuracy (%)
EAU-Net [15]	81.83
DeepExp3D [16]	82.06
ConvNeXt	82.60
pACNN [8]	83.27
Swin Transformer	83.50
MobileNetV2 [17]	83.54
Model of this paper	83.80

TABLE 5 | Comparison of the accuracy of different models on the FER2013 dataset.

Model	Accuracy (%)
AlexNet [9]	61.10
GoogleNet [18]	65.20
Parallel CNN [19]	65.60
VGGNet16 [11]	66.80
SHCNN [9]	69.10
MobileNetV2 [17]	69.49
ConvNeXt	69.60
ResNet18 [20]	69.75
Swin Transformer	70.20
Model of this paper	70.40

proposed in this paper is simultaneously validated on the Fer2013 dataset and compared with state-of-the-art models such as GoogleNet, AlexNet, VGGNet, MobileNet, and ResNet (see Table 5). Notably, the accuracy of the improved ConvNeXt network is 9.3%, 5.2%, and 3.6% higher than that of AlexNet, GoogleNet, and VGGNet16 networks, respectively. Moreover, compared with MobileNetV2, Shallow CNN, and other models, the accuracy of the proposed model has significant advantages.

In order to assess the stability of the improved model more intuitively, this paper records the model loss value during the training process, so as to monitor the training process of the model in real time. As shown in Figure 7.

4.3.2 | Ablation Experiments

To demonstrate the effectiveness of the SENET attention mechanism and FL function, we performed ablation experiments on the RAF-DB and Fer2013 datasets. The details are shown in Table 6. After incorporating the SENET attention mechanism and FL function, the accuracy of facial expression recognition improved by 1.2 and 0.8 percentage points on the RAF-DB and Fer2013 datasets, respectively. The results indicate that the introduced SENET attention mechanism can accurately identify key regions, enhance feature extraction, and suppress irrelevant features.

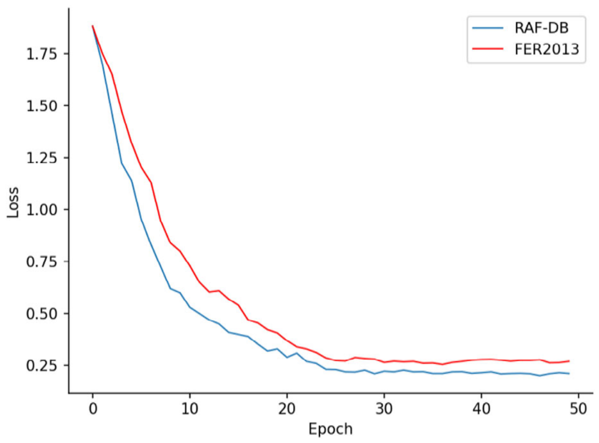

FIGURE 7 | Loss value curves.

TABLE 6 | The results of ablation experiment were compared.

SENET	FL	RAF-DB (%)	Fer2013 (%)
+	+	83.8	70.4
−	+	82.9	69.7
+	−	82.9	70.2
−	−	82.6	69.6

The FL effectively alleviates the problem of sample imbalance, reducing the loss to some extent. In conclusion, the introduction of the SENET attention mechanism and the FL function both help to improve the detection performance of the model.

5 | Conclusion

This study introduces an enhanced ConvNeXt network model for facial expression recognition, which incorporates the ConvNeXt-T network and SENET to detect seven basic facial expressions in natural settings. By leveraging the SENET module, the network learns the significance of channels, thereby emphasizing the extraction of key features. Moreover, the utilization of the Focal loss function helps mitigate sample distribution imbalances. The integration of SENET and FL in the ConvNeXt network proves to be more suitable for facial expression recognition and classification tasks, as evidenced by comparisons on the RAF-DB and Fer2013 datasets. Given the improved accuracy achieved by the ConvNeXt network, there are plans to integrate the model into intelligent robots to enhance their human-like and intelligent capabilities. Furthermore, the model is also intended to be utilized in recognizing emotions in autistic children, potentially aiding researchers and medical professionals in this domain.

However, the proposed method is still insufficient for solving face expression feature extraction under complex backgrounds, such as facial occlusion, light intensity, and pose deflection. In future work, the accuracy of the model is further improved by solving cases such as occluded facial expressions of faces under actual complex backgrounds.

Author Contributions

Dan Chen: writing – review and editing, investigation, conceptualization, validation, software. **Yu Cao:** writing – review and editing, writing – original draft, funding acquisition. **Xu Cheng:** writing-review and editing, methodology, project administration.

Acknowledgements

This work was supported in part by the Applied Basic Research Program of Liaoning Province under Grant 2022JH2/101300272, in part by the National Natural Science Foundation of China under Grant 71902121 and in part by the Applied Basic Research Program of Liaoning Province (Key Technology Research on Non-standard Agricultural Product Quality Grading Based on Artificial Intelligence Technology).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Baidu Netdisk at [https://pan.baidu.com/s/1kHCXLXR-KEDBuaWn5OmvTcA?pwd=qrjh], extraction code [qrjh].

References

1. C. Shan, S. Gong, and P. W. McOwan, "Facial Expression Recognition Based on Local Binary Patterns: A Comprehensive Study," *Image and Vision Computing* 27, no. 6 (2009): 803–816.

2. O. Déniz, G. Bueno, J. Salido, J. Salido, and F. De la Torre, "Face Recognition Using Histograms of Oriented Gradients," *Pattern Recognition Letters* 32, no. 12 (2011): 1598–1603.

3. C. C. Atabansi, T. Chen, R. Cao, and X. Xu, "Transfer Learning Technique With VGG-16 for Near-Infrared Facial Expression Recognition," *Journal of Physics: Conference Series* 1873, no. 1 (2021): 012033.

4. B. Li and D. Lima, "Facial Expression Recognition via ResNet-50," *International Journal of Cognitive Computing in Engineering* 2 (2021): 57–64.

5. J. Y. Lei, W. Ma, M. Xia, et al., "Research on Emotion Recognition of Face Image Based on densenet," *Journal of South-Central Minzu University (Natural Science Edition)* 42, no. 6 (2023): 781–787.

6. N. Sun, Q. Li, R. Huan, J. Liu, and G. Han, "Deep Spatial-temporal Feature Fusion for Facial Expression Recognition in Static Images," *Pattern Recognition Letters* 119 (2019): 49–61.

7. Y. Nan, J. Ju, Q. Hua, H. Zhang, and B. Wang, "A-Mobilenet: An Approach of Facial Expression Recognition," *Alexandria Engineering Journal* 61, no. 6 (2022): 4435–4444.

8. Y. Li, J. Zeng, S. Shan, and X. Chen, "Occlusion Aware Facial Expression Recognition Using CNN With Attention Mechanism," *IEEE Transactions on Image Processing* 28, no. 5 (2018): 2439–2450.

9. S. Miao, H. Xu, Z. Han, and Y. Zhu, "Recognizing Facial Expressions Using a Shallow Convolutional Neural Network," *IEEE Access* 7 (2019): 78000–78011.

10. M. Bie, H. Xu, Y. Cao, K. Song, and X. Che, "Swin-FER: Swin Transformer Facial Expression Recognition," *Applied Sciences* 14, no. 14 (2024): 6125.

11. H. Nie, "Face Expression Classification Using Squeeze-Excitation Based VGG16 Network," in *Proceedings of the 2022 2nd International Conference on Consumer Electronics and Computer Engineering (ICCECE)* (2022): 482–485.

12. K. Zhang, Q. Liu, H. Qian, et al., "EATN: An Efficient Adaptive Transfer Network for Aspect-Level Sentiment Analysis," *IEEE Transactions on Knowledge and Data Engineering* 35, no. 1 (2021): 377–389.

13. Y. Yu, M. Wan, J. Qian, D. Miao, Z. Zhang, and P. Zhao, "Feature Selection for Multi-Label Learning Based on Variable-Degree Multi-Granulation Decision-Theoretic Rough Sets," *International Journal of Approximate Reasoning* 169 (2024): 109181.

14. X. Li, D. Li, P. Yuan, et al., "Fusion Innovation: Multi-Scale Dilated Collaborative Model of ConvNeXt and MSDA for Fault Diagnosis," *Robotics and Autonomous Systems* 182 (2024): 104819.

15. J. Deng, G. Pang, Z. Zhang, Z. Pang, H. Yang, and G. Yang, "cGAN Based Facial Expression Recognition for human-Robot Interaction," *IEEE Access* 7 (2019): 9848–9859.

16. M. R. Koujan, L. Alharbawee, G. Giannakakis, N. Pugeault, and A. Roussos, "Real-Time Facial Expression Recognition 'in the Wild' by Disentangling 3d Expression From Identity," *Proceedings of the 15th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2020)* (2020): 24–31.

17. X. Y. Deng, H. Y. Pei, and Y. Sheng, "Facial Expression Recognition Based on Network Fusion to Improve MobileViT," *Computer Engineering and Science* 46, no. 6 (2024): 1072–1080.

18. Y. E. I. Boudouri and A. Bohi, "EmoNeXt: An Adapted ConvNeXt for Facial Emotion Recognition," in *Proceedings of the 2023 IEEE 25th International Workshop on Multimedia Signal Processing (NMSP)* (2023): 27–29.

19. L. L. Xu, S. M. Zhang, and J. L. Zhao, "Expression Recognition Algorithm for Parallel Convolutional Neural Networks," *Journal of Image and Graphics* 24, no. 2 (2019): 227–236.

20. X. Zhao, C. Yang, R. N. Wang, and Y. C. Li, "Facial Expression Recognition Based on Attention Mechanism ResNet Lightweight Network," *Chinese Journal of Liquid Crystals and Displays* 38, no. 11 (2023): 1503–1510.